

Early forecasting of corn yield and price variations using satellite vegetation products

Florian Teste^{a,b,*}, David Makowski^{a,**}, Hassan Bazzi^{a,b}, Philippe Ciaï^c

^a Université Paris-Saclay, INRAE, AgroParisTech, UMR MIA 518, 91120, Palaiseau, France

^b Atos France, Technical Services, 95870, Bezons, France

^c CEA-CNRS-UVSQ, Université Paris-Saclay, IPSL, LSCE, Gif-sur-Yvette, 91190, France

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ABSTRACT

Several approaches based on process-based, statistical or economic models have been developed and are currently used by market analysts and governmental agencies to forecast agricultural yields and commodity prices at the national and global scales. However, these models require input data that are often difficult to obtain before crop harvest and their predictions are not always reliable. Satellite data products provide easily accessible sources of information on vegetation characteristics, potentially relevant for predicting crop yield and commodity prices at a large scale, before harvest. This study presents and evaluates several methods for forecasting variations in US corn yield and global corn prices using regional vegetation satellite products. The spatial dimension of maps of vegetation indices derived from satellite products in the US Corn Belt is first reduced either by their spatial means or by principal components derived using the Empirical Orthogonal Function (EOF) method. Mean values and EOF principal components are then used as explanatory variables in generalized LASSO linear models to forecast corn price and yield changes at different time horizons, before harvest. A cross-validation method was implemented to assess the model performances using Area under the ROC curve (AUC), the Brier Skill Score (BSS) and the Matthew Correlation Coefficient (MCC). The results revealed that EOF principal components derived from satellite-based gross primary production (GPP) in July could be used to predict variations in US corn yields, with an AUC of 0.97 (95% CI 0.92–1), a BSS of 0.75 and an MCC of 0.83. The results also showed that the July spatial averages of satellite GPP data obtained in the US Corn Belt could be used to forecast the forthcoming increase or decrease of global corn prices as evidenced by an AUC reaching up to 0.91, a BSS up to 0.5, and an MCC up to 0.67. Finally, we compared our forecasts of yield and price variations to those derived from the WASDE forecasting method currently implemented in the US. This comparison showed that our approach was at least as accurate and often more accurate than WASDE. This study demonstrates how satellite data can be used to forecast variations in crop yields and corn prices several months before harvest.

1. Introduction

The demand for agricultural commodities is steadily rising, driven by population growth and changing dietary preferences. The Food and Agriculture Organization of the United Nations (FAO) predicts that the global population will reach 9.7 billion by 2050, requiring a significant increase in food production to meet this growing demand. According to recent estimates by van Dijk et al. (2021), the demand for food may increase by as much as 62% in 2050 compared to 2010, while Tilman et al. (2011) project a potential increase of 100% to 110% in crop demand between 2005 and 2050. With the looming threat of climate

change, ensuring global food security has emerged as one of the most critical challenges of the 21st century (Zhao et al., 2017; Jägermeyr et al., 2021). Several studies have already highlighted the significant impact of climate change on crop production, particularly for staple crops such as corn, wheat, and rice, indicating that the supply of these crops may be at risk. It is estimated that for each degree of global temperature increase, wheat yields could reduce by 4%–6%, while temperature increases could similarly affect maize yield the rice and soybeans by 3% (Zhao et al., 2017; Asseng et al., 2015; Liu et al., 2016).

Crop yields and commodity prices are intertwined. Zelingher et al. (2021), Zelingher and Makowski (2022) have demonstrated that yearly

* Corresponding author at: Université Paris-Saclay, INRAE, AgroParisTech, UMR MIA 518, 91120, Palaiseau, France.

** Corresponding author.

E-mail addresses: florian.teste@agroparistech.fr (F. Teste), david.makowski@inrae.fr (D. Makowski), Hassan.bazzi@atos.net (H. Bazzi), philippe.ciais@lsce.ipsl.fr (P. Ciaï).

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crop production variations strongly influence agricultural commodity prices. Specifically, these studies have shown that variation in US corn yields significantly impacts global corn prices, underscoring the critical importance of accurate yield forecasts in this influential region. Assessing the risk of shocks in agricultural yields and commodity prices at the national or global scales is of interest to a wide range of stakeholders, including policymakers, traders, the food industry, and consumers. Because of the strong relationship between global and local prices (Headey and Fan, 2010), unexpected production losses can have significant implications for food security (Laborde et al., 2021). Mundlak and Larson (1992) showed that adjustments in global prices are typically mirrored in household (consumer) prices, thereby directly impacting consumers. Consequently, anticipating such production and price shocks is crucial to safeguard food security proactively. Large-scale yield and price forecasts provide a baseline of expected crop production and food balance, widely used to monitor the food market and trigger policy response for food security (Declaration, 2011). Yield and price forecasting tools could facilitate the timely activation of mitigation measures to minimize consumer risks and improve food security levels.

Several approaches have been applied for predicting crop yields before harvest at a large scale. On-farm surveys are powerful tools to estimate crop yields based on field observations (Doré et al., 1997), but they are time-consuming and costly. Alternatively, process-based crop models are able to forecast crop yields from a set of equations describing radiation interception, biomass production, water balance, and nutrient uptake (Lobell and Asseng, 2017). These models include numerous inputs related to climate and soil characteristics and agricultural practices. Although these models can provide valuable insights into the underlying processes that govern crop growth and yield, their accuracy is being questioned as they cannot consistently reproduce major yield loss events and do not always perform better than simple weather indices.

The prediction of agricultural commodity prices has also been the subject of extensive work. Most of these efforts rely on time series forecasting models such as the autoregressive moving average (ARIMA) and its extensions, as presented by Adebiyi et al. (2014), Kohzadi et al. (1996). These models predict future price values based on historical values using autocorrelation in time series data. Machine learning models have recently been implemented for agricultural commodity price forecasting. Studies by Zelingher et al. (2021), Zelingher and Makowski (2022), Xu and Zhang (2022) demonstrate the potential of machine learning algorithms to provide accurate predictions and highlight the advantages of such models, including their ability to capture complex patterns in the data. Some of these models Zelingher et al. (2021), Zelingher and Makowski (2022) are able to predict yearly price variations from yearly crop yield (or production) data. However, reliable information on crop yields is not always easily accessible, especially when the objective is to predict price variation before harvest. This poses a challenge for price modeling, as inaccurate or incomplete data on crop yields can lead to errors in forecasting supply and demand.

In contrast, remote sensing is a highly effective means of monitoring large agricultural areas, providing precise and timely information. Its ability to collect data over large areas with frequent revisits makes it particularly well-suited for this task. Remote sensing sensors collect detailed information on crop spectral properties at different crop development stages, which can be used to compute various spectral indices, such as the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI). Additionally, biophysical parameters, such as Leaf Area Index (LAI), Fraction of Absorbed Photosynthetically Active Radiation (FAPAR), and Gross Primary Production (GPP), can be estimated using the data collected (Liang et al., 2021; Tramontana et al., 2016). These vegetation indices and biophysical parameters are valuable for assessing crop growth, predicting yield and production before harvest at a large scale, and identifying stress factors, such as water or nutrient deficiencies (Becker-Reshef et al., 2010; Wardlow

and Egbert, 2008; Johnson, 2014). By providing information on crop production, remote sensing data could be used to forecast commodity price changes. Still, such an approach has not been attempted to date.

Our objective is to define, implement and test a new methodological framework to forecast yearly variations of crop yields and commodity prices at a large scale from publicly available remote sensing data. The proposed framework does not use any climate, soil, or crop characteristics data and relies only on vegetation indices derived from remote sensing data collected before harvest. It can thus be easily used to predict yield and price changes a few months before harvest. The value of this framework is demonstrated for corn production in the US, a prominent and influential corn-producing area for the global price.

2. Materials & methods

2.1. Data

2.1.1. Corn price

We obtained global monthly prices for maize (US No. 2 yellow free on board (FOB) Gulf of Mexico) from the World Bank's commodity markets database, measured in US nominal price per metric tonne; the historical time series cover the period from 1960 to 2021. This study represents the monthly corn price by p_{mt} , where m is the month, and t is the year. Then, based on a year-to-year increment, the relative difference in price compared to the same month in the previous year is expressed as:

$$\Delta p_{mt} = \frac{p_{mt} - p_{mt-1}}{p_{mt-1}} \quad (1)$$

Additionally, we defined a binary variable, Δp_{mt}^b equal to one in case of price increase ($\Delta p_{mt}^b = 1$ when $\Delta p_{mt} > 0$) and to zero otherwise. Different variables Δp_{mt} and Δp_{mt}^b were defined for the months of March to December.

2.1.2. Corn yield

The United States, as the world's leading corn producer, accounts for a significant percentage of global corn production, at 30% in 2022 (the market year 2022/2023). A large portion of this production, approximately 80%, is concentrated in the US Corn Belt region, making it a suitable area for our study, which encompasses thirteen Midwestern states, including Illinois (IL), Indiana (IN), Iowa (IA), Kansas (KS), Kentucky (KY), Michigan (MI), Minnesota (MN), Missouri (MO), Nebraska (NE), North Dakota (ND), Ohio (OH), South Dakota (SD), Wisconsin (WI). We obtained historical data on annual US corn yields (in hectograms per hectare) from 1961 to 2021 from the FAOSTAT.

In our study, the annual corn yield in the USA at time t is represented by $yield_t$. To analyze the yearly yield variations, we calculated the relative yield difference between consecutive years, noted $\Delta yield_t$, as follows:

$$\Delta yield_t = \frac{yield_t - yield_{t-1}}{yield_{t-1}} \quad (2)$$

Based on these relative differences, we also defined a binary variable, $\Delta yield_t^b$, equal to one in case of a yield increase ($\Delta yield_t > 0$) and zero otherwise.

2.1.3. Corn mask

The Corn Belt is one of the world's most intensive agricultural production areas. In this region, corn (*Zea mays* L.) and soybean (*Glycine max* (L.) Merr.) account for about 75% of the agricultural land NASS (2022), Roesch-McNally et al. (2018). To specifically focus our study on the corn-producing area, we applied a corn mask using the USDA National Frequency Layer (2021). This layer, derived from land cover information from 2008 to 2021 (Cropland Data Layer) (USDA-NASS, 2022), identifies crop-specific planting frequency and is made available in a raster format with a resolution of 30 m. From this raster layer, we first removed non-corn pixels and then in a second step, we removed pixels where corn was not grown for at least five years during the 2008–2021 period. We finally aggregated the pixels at a spatial resolution of 0.05°.

2.1.4. Remote sensing data and vegetation indices

Remote sensing technologies have transformed the modern agricultural analytical landscape. During the last decade, the abundance of sensors and platforms have led to the accumulation of large datasets on a global scale. Several vegetation indices derived from remote sensing data were obtained from the Global Land Surface Satellite (GLASS) program, including the Leaf Area Index (LAI), Fraction of Absorbed Active Radiation (FAPAR), and Gross Primary Products (GPP). These satellite products provide information on plant cover characteristics encompassing the effects of farmers' practices (Nie et al., 2022) and environmental factors on crop growth. Compared to other sources, the GLASS data stands out for its ability to capture vegetation dynamics on a global scale, as it has an extended temporal coverage starting from 1982 to 2018 and provides images every eight days at a spatial resolution of 0.05°. We aggregated the GLASS LAI, FAPAR and GPP products on a monthly basis, from February to October (the month of corn harvest). The three vegetation indices are further denoted as z_{smt}^v , where s is the pixel, m is the month, t the year and v the type of products (LAI, FAPAR, GPP). The high-dimensional characteristic of remote sensing is a topic of high interest, as evidenced by the increasing number of publications on the subject (Bioucas-Dias et al., 2013; Li et al., 2022). In our case, the number of pixels s ($\approx 163,000$ pixels) is much greater than the number of temporal dimensions t (37 years of observations). Note that the volume of data used in our study is not particularly large by today's standards. We use more than 5100 images, but this represents 30.3 GB only. We present below several strategies to reduce the spatial dimension of the vegetation indices. We then present several models to predict yield and price variations from the vegetation indices.

2.2. Methods

2.2.1. Dimension reduction

We used two approaches to reduce the number of pixels. The first one was very simple and based on the spatial average of vegetation indices. The second one was based on Empirical Orthogonal Function (EOF).

The spatial average of z_{smt}^v for each month and each year was defined as follows:

$$\bar{z}_{mt}^v = \frac{1}{S} \left(\sum_{s=1}^S z_{smt}^v \right) \quad (3)$$

For a given vegetation index v (FAPAR, LAI, GPP), each average value represents the spatial average of the index values for a given month and a given year.

We computed a second type of average vegetation index based on the yearly difference between year t and year $t-1$. For a given vegetation index v and a given month m , the yearly difference Δz_{smt}^v was defined as :

$$\Delta z_{smt}^v = \frac{z_{smt}^v - z_{smt-1}^v}{z_{smt-1}^v} \quad (4)$$

Then, the spatial average of the yearly difference was computed as:

$$\Delta \bar{z}_{mt}^v = \frac{1}{S} \left(\sum_{s=1}^S \Delta z_{smt}^v \right) \quad (5)$$

The second approach is based on the Empirical Orthogonal Function (EOF) analysis, a widespread technique in climate science dating back to the 1950s (Hannachi et al., 2007). EOF is a popular decomposition technique used to depict the spatial and temporal variations of time series data based on a small number of variables (principal components).

The EOF decomposition of a spatiotemporal matrix $\mathbf{X}$ with the temporal dimension t as rows and the number of pixels s as columns is

achieved using the singular value decomposition (SVD) method (Hannachi et al., 2007; Hannachi, 2004) such as :

$$\mathbf{X} = \mathbf{A} \mathbf{\Lambda} \mathbf{U}^T, \quad x_{ts} = \sum_{k=1}^E a_{kt} \lambda_k u_{ks}, \quad (6)$$

The SVD decomposition splits the matrix $\mathbf{X}$ into three constituent matrices: an orthogonal $T \times T$ matrix (T is the total number of years), $\mathbf{A}$, containing the principal components (PCs), another orthogonal $S \times S$ matrix (S is the total number of pixels), $\mathbf{U}^T$, containing the EOFs (spatial pattern) and a diagonal matrix $\mathbf{\Lambda}$ containing the singular values. The orthogonal matrices, $\mathbf{A}$ and $\mathbf{U}^T$, are independent representations of the temporal and spatial variations in the original datasets. The first EOF is the mode (i.e., linear combination of the S pixels) that explains the highest fraction of variance in the data, and the second EOF is the mode that explains the second-highest fraction of variance, and so on. The total number of EOFs, E , corresponds to the smallest value among the number of rows and the number of columns, $E = \min(T, S)$. Jolliffe (2002) and Wilks (2006) provide a comprehensive description of the underlying EOF/PCA principles and methods.

The EOF-SVD decomposition was applied month-by-month to the vegetation indices z_{smt}^v and Δz_{smt}^v , successively, using the package wq of R (Jassby and Cloern, 2022). For a given month m , $\mathbf{X}$ contains either the $T \times S$ values of z_{smt}^v or the $T \times S$ values of Δz_{smt}^v , and the EOFs are the linear combinations of pixels explaining the most the variability of the z_{smt}^v and Δz_{smt}^v . The principal component associated to the k th EOF ($k = 1, \dots, E$) is a vector including T values, noted a_{kt} , $t = 1, \dots, T$. Fig. 1 illustrates the output obtained from the Empirical Orthogonal Function-Singular Value Decomposition (EOF-SVD) analysis. Specifically, panels A and B display the first and second EOFs, which account for 46.1% and 5.7% of the variance, respectively. Panels C and D present the corresponding principal components for each EOF, indicating the temporal fluctuation of the vegetation index (here, July annual variation of GPP).

2.2.2. Models

The principal components (a_{kt} , $k = 1, \dots, E$) derived from the EOF decomposition, along with the spatial averages ($\bar{z}_{mt}^v$ and $\Delta \bar{z}_{mt}^v$), were used as predictors in generalized linear models (GLM) to estimate the probabilities of yield and price increase compared to the previous year.

The models computing the probabilities of price and yield increase from the EOF-principal components were defined as follows:

$$\Pr(\Delta p_{mt}^b = 1 \mid a_{1t}, \dots, a_{kt}, \dots, a_{Et}) = \frac{e^{\beta_0 + \sum_{k=1}^E \beta_k a_{kt}}}{1 + e^{\beta_0 + \sum_{k=1}^E \beta_k a_{kt}}} \quad (7)$$

$$\Pr(\Delta \text{yield}_t^b = 1 \mid a_{1t}, \dots, a_{kt}, \dots, a_{Et}) = \frac{e^{\beta_0 + \sum_{k=1}^E \beta_k a_{kt}}}{1 + e^{\beta_0 + \sum_{k=1}^E \beta_k a_{kt}}} \quad (8)$$

The models computing the probabilities of price and yield increase from the spatial averages of vegetation indices ($\bar{z}_{mt}^v$) were defined as follows:

$$\Pr(\Delta p_{mt}^b = 1 \mid \bar{z}_{mt}^v) = \frac{e^{\beta_0 + \beta_1 \bar{z}_{mt}^v}}{1 + e^{\beta_0 + \beta_1 \bar{z}_{mt}^v}} \quad (9)$$

$$\Pr(\Delta \text{yield}_t^b = 1 \mid \bar{z}_{mt}^v) = \frac{e^{\beta_0 + \beta_1 \bar{z}_{mt}^v}}{1 + e^{\beta_0 + \beta_1 \bar{z}_{mt}^v}} \quad (10)$$

The models computing price and yield increase probabilities from the spatial average of the yearly difference of vegetation indices ($\Delta \bar{z}_{mt}^v$) were defined as follows:

$$\Pr(\Delta p_{mt}^b = 1 \mid \Delta \bar{z}_{mt}^v) = \frac{e^{\beta_0 + \beta_1 \Delta \bar{z}_{mt}^v}}{1 + e^{\beta_0 + \beta_1 \Delta \bar{z}_{mt}^v}} \quad (11)$$

$$\Pr(\Delta \text{yield}_t^b = 1 \mid \Delta \bar{z}_{mt}^v) = \frac{e^{\beta_0 + \beta_1 \Delta \bar{z}_{mt}^v}}{1 + e^{\beta_0 + \beta_1 \Delta \bar{z}_{mt}^v}} \quad (12)$$

We developed a specific model for each type of vegetation index (LAI, fAPAR, GPP) and each month of observation (from February to October). Vegetation indices have not been cumulated over previous

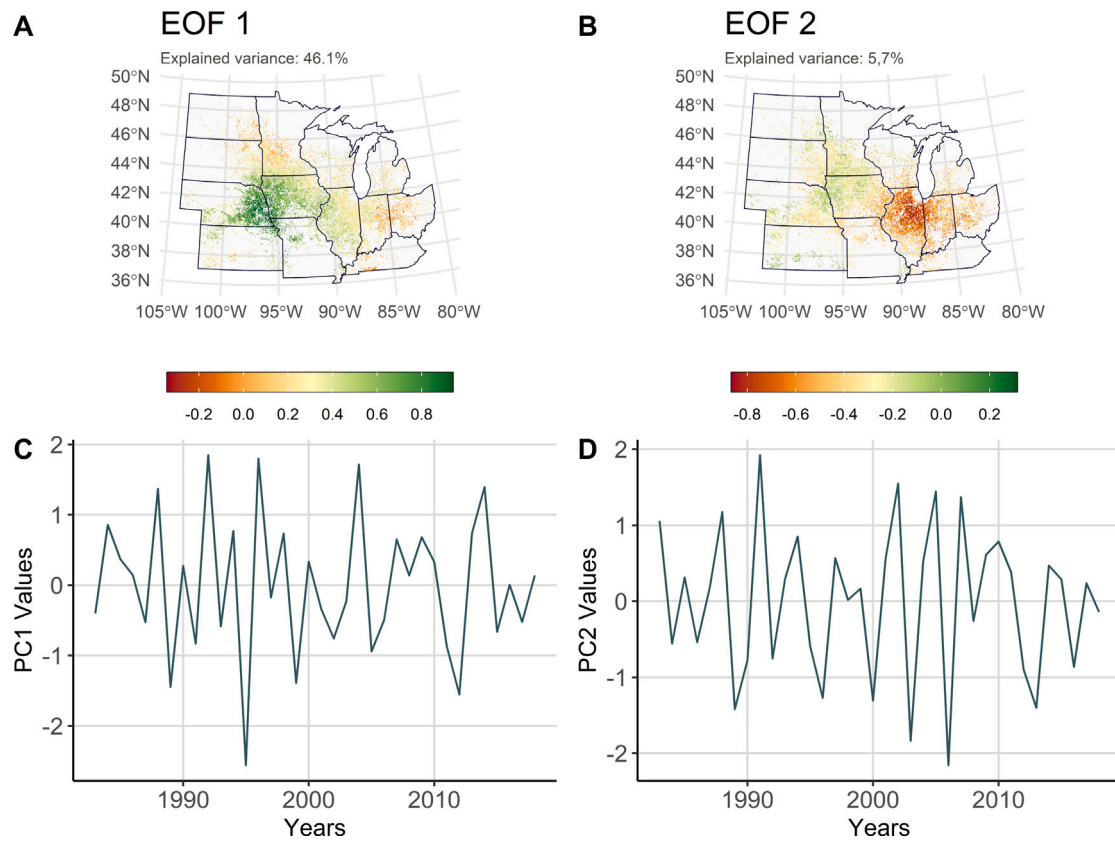


Fig. 1. Panel A presents the first EOF from the EOF-SVD decomposition of July's annual GPP variation, accounting for 46.7% of the dataset variance. Panel C shows the corresponding first principal component, capturing the temporal dynamics of this spatial pattern. Panel B and D display the second EOF and its principal component respectively, explaining the largest share of the variance (5.7%) unaccounted by the first EOF.

months, as the vegetation characteristics observed in a given month result from all the events that have impacted the plant cover from sowing to the date (month) of observation. All models were fitted using the GLM function of R (R Core Team, 2022). With the models using principal components as predictors, the most relevant principal components were selected using the LASSO (Least Absolute Shrinkage and Selection Operator) method implemented with the GLMnet package in R (Simon et al., 2011). The number of yield data available is intrinsically limited by the fact that only one yield data can be collected every year (because only one corn crop is cultivated every year). Similarly, only one yearly price change data at a given month (e.g., October price change compared to the previous year same month) is available every year. We thus selected a model type that could effectively capture the underlying patterns of the data with a low risk of overfitting. Due to their simplicity and parsimony, generalized linear models (GLM) with LASSO regularization are a natural choice and have proven their effectiveness in similar contexts (Sharif et al., 2017). Moreover, thanks to its penalty term, GLM LASSO is able to select the most influential input variables in predicting yield and price changes, which makes it easier to interpret the results.

2.2.3. Model evaluation

Model performances were evaluated by using the leave-one-year-out Cross-Validation (LOOCV) procedure. The first step is to divide the entire T data set into two parts: the test data set, which includes the year t chosen for testing, and the training data set, which consists of the remaining $T - 1$ years. After splitting the data, the models are fitted to the training data set. The fitted models are then applied to the test dataset including the year t and the corresponding outputs are computed. We repeat these steps T times to generate the model outputs for all years. These outputs are used to evaluate the model

performances by computing the Area Under the Curve (AUC), the Brier Skill Score (BSS) and the Matthew correlation coefficient (MCC). The AUC values of perfect and useless models are equal to one and 0.5, respectively. A BSS equals to one means that the model makes perfect predictions. A BSS of zero indicates no improvement over the reference model, while a negative BSS indicates that the model has degraded performance compared to the reference model. Here, the reference model is a GLM model without any predictor. The AUC provides information on the capabilities of the models to classify yield/price increase/decrease correctly. The BSS measures the distance between the probabilities computed by the models and the observed binary variables indicating yield/price increase/decrease.

The MCC is a metric commonly used to assess binary classifiers. It has proved to be particularly useful in the context of imbalanced datasets. Unlike other measures, MCC incorporates information on true positives, true negatives, false positives, and false negatives through a contingency matrix format. MCC is calculated as the Pearson product-moment correlation coefficient between predicted and actual binary classifications, here the actual and predicted yield increase/decrease events (Powers, 2020). It is expressed as follows:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}} \quad (13)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

As mentioned by Chicco (2017), the MCC is a binary classification metric taking a high score solely when the binary predictor accurately forecasts the majority of positive and negative data instances. The metric takes values within an interval ranging from -1 to 1 , with -1 and 1 representing perfect misclassification and perfect classification, respectively. An MCC equal to zero indicates that the classifier performance is no better than a random guess.

2.2.4. Comparison with the WASDE predictions

The WASDE (World Agricultural Supply and Demand Estimates) report is a crucial monthly publication of the United States Department of Agriculture (USDA) that provides an assessment of current and projected supply and demand for major agricultural commodities in the U.S. and around the world (Vogel and Bange, 1999). The ever-changing dynamics of the global market for agricultural commodities necessitate a reliable source of information, and the WASDE report serves as a crucial resource in this regard. As a significant element of the public information system in agriculture, key stakeholders frequently use the report, including farmers, traders, and policymakers (Isengildina-Massa et al., 2008a,b).

Here, we focus on a specific element in the WASDE reports: the July estimation of US corn yield from 1993 (when WASDE first made available corn yield and harvested area estimates) to 2018. In order to compare the WASDE's quantitative predictions of corn yield with our predictions of yield increase or decrease, we converted the data into binary indicators. We employed the method described in Eq. (2) to calculate the yield variation presented in the WASDE reports. We then created a binary variable, which takes on a value of one if the WASDE report predicted an increase in corn yield and zero otherwise.

We evaluated the performance of the WASDE-based binary predictions and the yield models using the same metrics: the area under the curve (AUC), the Brier skill score (BSS) and the Matthews correlation coefficient (MCC). We applied Leave-One-Year-Out cross-validation to calculate these metrics and used 0.5 as a probability threshold for classification.

3. Results & discussion

3.1. Yield variations

Fig. 2 shows the values of BSS obtained with monthly Gross Primary Productivity (GPP) and yearly differences of monthly GPP (month of one year compared to the same month the previous year). The values of BSS are always close to zero or even negative between February and May except in April, showing that the models are mostly useless before sowing in Spring and at the beginning of the growing season. However, values of BSS become positive in June and peak in July when they exceed 0.5. Then, BSS values tend to decline as we approach the harvest season in October. The highest overall performance is achieved using empirical orthogonal function (EOF) predictors obtained from yearly GPP difference, with a BSS of 0.75 in July. A key finding is that the model's performance improves notably when using predictors obtained from yearly GPP differences rather than those directly derived from the GPP Fig. 2. This is evidenced by significantly higher Brier skill score (BSS) values – with a BSS of 0.75 for EOF predictors derived from yearly GPP differences compared to a negative BSS for GPP-based EOF predictors in July and 0.46 compared to negative BSS in August – and by the superior performance of GPP yearly differences compared to the barely positive BSS of the GPP products.

Figs. A.1, A.2, and A.3 provide further details on the BSS scores for the different predictors. The Brier Skill Score (BSS) values for predictions made using the Leaf Area Index (LAI) and Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) products are significantly lower compared to those obtained with Gross Primary Productivity (GPP). Interestingly, the LAI and FAPAR yearly differences shift from negative in the first half of the year to positive from July through September. In particular, the yearly LAI differences peak at 0.12 in July, then decrease to 0.11 in August and September, while the FAPAR yearly differences reach their maximum value at 0.14 in July, then gradually decrease to 0.13 in August and 0.12 in September. These low BSS values show that models utilizing FAPAR or LAI products have lower performances than those based on GPP products.

Another noteworthy finding from the study is that the first EOF derived from yearly GPP differences (selected through LASSO and displayed in Table A.1) is associated with the largest estimated regression

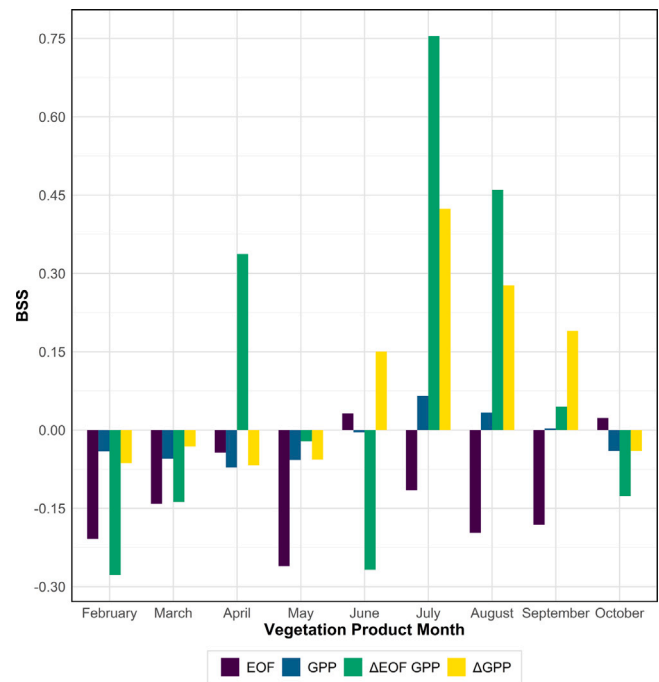


Fig. 2. Brier Skill Score (BSS) assessing predictions of US corn yield annual variation obtained at different dates, from February to October (start of the harvest period). EOF: predictions based on EOF applied to monthly Gross Primary Production (GPP) available from February to October. GPP: predictions based on spatial averages of monthly GPP computed from February to October. ΔEOF GPP: predictions derived from EOF applied to yearly differences of monthly GPP (month of one year compared to the same month of the previous year). ΔGPP: predictions derived on spatial averages of yearly differences of monthly GPP.

coefficient in the model ($\beta_k = 3.69$). The first EOF captures 46.1% of the variance and manifests positive loadings in Eastern Nebraska and Western Iowa pixels. In contrast, it exhibits negative loadings in Indiana and Ohio. As these areas are all crucial corn-producing regions, it is logical that they substantially influence yearly variations in US corn yield.

This finding reveals a direct and positive relationship between the first EOF and the probability of an increase in US corn yield. In other words, the higher the principal component of the first EOF, the higher the chance of having a US corn yield higher than the previous year's yield.

The model with the highest BSS shows good classification performance, as illustrated in Fig. 3. The boxplot indicates that the predicted probabilities of yield increase (obtained by leave-one-year-out cross-validation) tend to be substantially higher for the years when a US yield increase is actually observed. The median predicted probability of yield increase is close to 1 at 0.98 in the case of actual yield increase and slightly above 0.067 in the case of actual yield decrease. The good classification performance is confirmed by the Receiver Operating Characteristic (ROC) curve shown in Fig. 3.B, characterized by an Area Under the Curve (AUC) of 0.97 with a 95% Confidence Interval of [0.92-1] (an AUC of 0.5 would indicate that the model performs no better than a random guess). The MCC scores of 0.83 indicate a strong correlation between the predicted and the observed yield variation.

The scatter plots provide an additional visual representation of this relationship between the predicted probabilities of yield increase and the actual yield variations. When the model predicts a probability of an increase greater than 0.5, most actual yield changes are greater than 0, indicating a yield increase. Conversely, when the predicted increase probabilities are lower than 0.5, most of the actual yield changes are lower than 0, indicating a yield decrease.

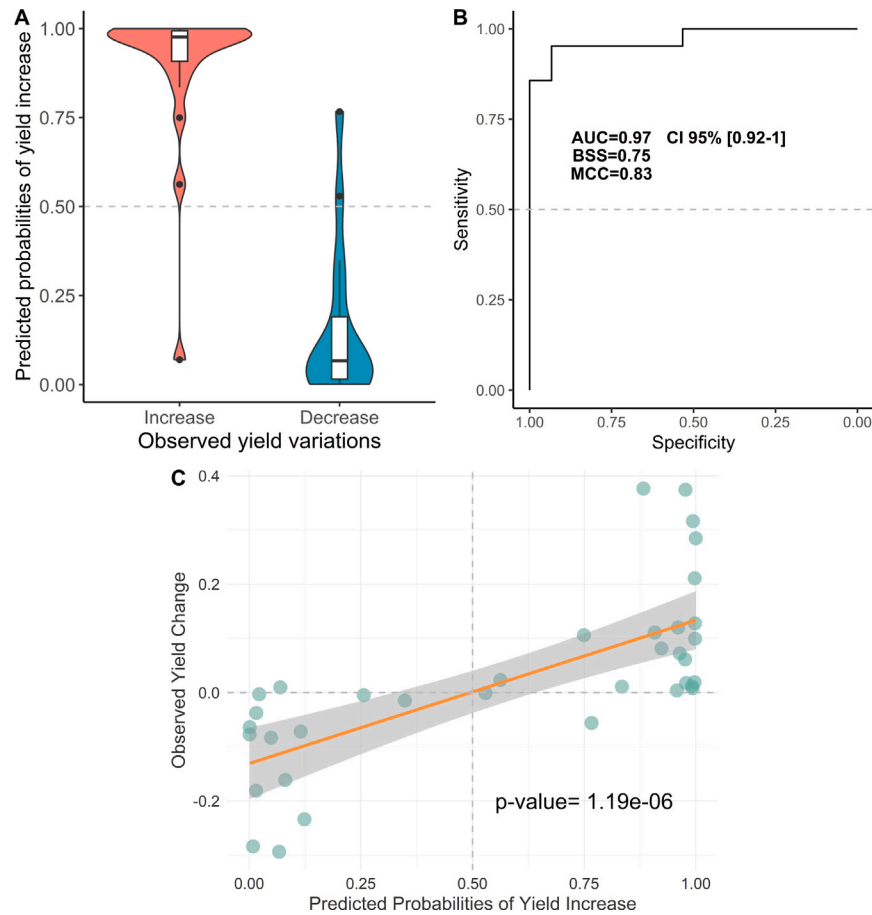


Fig. 3. Boxplots (A) shows the predicted probabilities of U.S. yield increase obtained with the best GLM-LASSO model for years of actual yield increase and decrease, separately. The model accuracy and correlation with actual yield variations are shown by the ROC curve (B) and MCC scores. The scatter plot (C) shows the relationship between predicted probabilities and actual yield changes, with probabilities above 0.5 mostly indicating yield increase and those below 0.5 implying yield decrease.

Interestingly, our model assessment shows that the best GLM model surpasses the WASDE report in predicting U.S. yield variations, as shown in Table 1. The WASDE yield prediction only achieved an MCC of 0.16 (where an MCC of 0 indicates a random guess) in predicting yield changes between 1993 and 2018. Additionally, the WASDE report had a nearly null BSS score of 0.003 and an AUC of 0.49, i.e. a bit lower than the 0.5 value characterizing a random classification. The AUC and BSS scores suggest that the best GLM (based on the EOFs of GPP yearly difference) has better classification performance. It achieved a substantially higher MCC of 0.43 compared to the WASDE approach, suggesting a robust correlation between predicted and observed yearly yield changes. The model also achieved a BSS of 0.18 and AUC of 0.75 over 1993–2018, both revealing a better classification performance than WASDE.

Fig. 4 demonstrates the poor performance of the yield classification based on the WASDE report. Specifically, as shown in Fig. 4.A, the WASDE model fails to distinguish between yield increase and decrease events, as there is very little difference in the predicted probabilities of yield increase from the WASDE reports between the years of actual yield increase and decrease. Even though Fig. 4.B indicates that the reported yield differences in the WASDE July reports are correlated with the actual yearly yield differences, this correlation alone is evidently insufficient to ensure accurate classification.

Weather significantly impacts global crop yield variability, contributing to about one-third of historical fluctuations (Ray et al., 2015). Severe weather events can negatively impact crop production, and

Table 1

Performance metrics of the models based on the World Agricultural Supply and Demand Estimates (WASDE) and the GLM using the Empirical Orthogonal Function (EOF) derived from the yearly variation of the July Gross Primary Productivity (GPP). The two approaches are compared using yield changes observed between 1993 and 2018.

BSS: Brier Skill Score; AUC: Area Under the Curve; MCC: Matthew Correlation Coefficient.

Model	BSS	AUC	MCC
WASDE	0.003	0.49	0.16
EOF ΔGPP	0.18	0.75	0.43

several studies consistently indicate that excessive heat, drought, and extreme wet and cold conditions lead to reduced crop yields (Li et al., 2019). As the satellite products considered in our study provide direct information on the characteristics of the vegetation cover, they can capture the effect of extreme weather conditions on plant growth. This is especially true for GPP, which inherently accounts for several climatic variables like temperatures. Nevertheless, the extreme events occurring after the dates of satellite observation cannot be accounted for and contribute to the prediction errors of our algorithm. This problem is not specific to our approach but is expected to all forecasting tools designed to predict crop yields before harvest. If reliable forecasts of future weather conditions are available, they could in principle, be easily incorporated as additional input variables into the LASSO

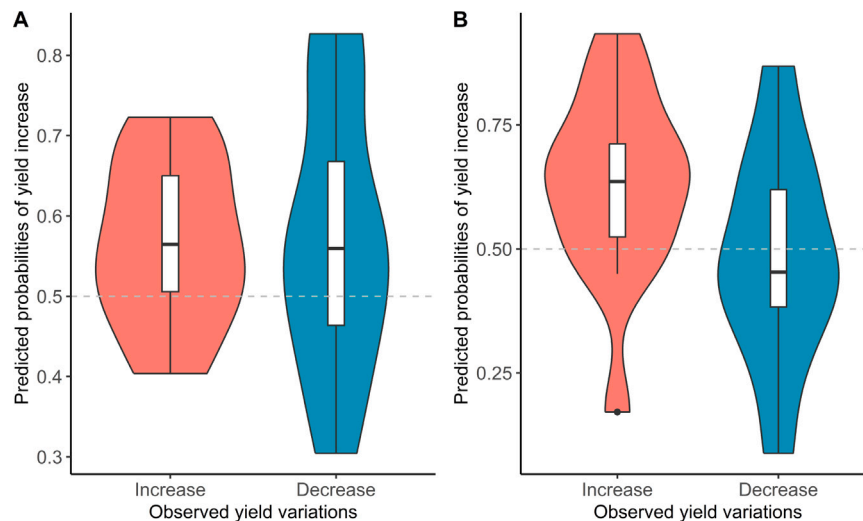


Fig. 4. Boxplot A presents the probabilities of yield variations predicted by a GLM model using the forecasts of relative yield change reported by the WASDE July reports as the predictor for the period 1993–2018, differentiating the years of actual yield increase and decrease. Boxplot B presents the probabilities of yield variations predicted by the GLM model, including the EOFs derived from the July GPP yearly differences as the predictor for the period 1993–2018, differentiating the years of actual yield increase and decrease.

regression used in this study to make the predictions. It would be interesting to test such an approach in a future study.

3.2. Price variations

The Brier Skill Scores used to assess the ability of the different remote sensing products to predict the corn price variation can be visualized in Figs. 5 & 7 and Appendices A.4 to A.15.

As for yield, model performances are lower when predictions are derived from FAPAR and LAI than with GPP. The GLM models using the spatial average of the yearly difference of July's FAPAR yield Brier Skill Scores (BSS) of 0.12, 0.26, 0.27, 0.25, and 0.2 for predicting corn price variations from August to December. These scores are comparable to those achieved by the spatial average of the yearly difference of July's LAI, which are 0.09, 0.25, 0.25, 0.25, and 0.19. The best performances are obtained with the spatial average of the annual difference of July's GPP, with BSS reaching 0.12, 0.38, 0.5, 0.31, and 0.17 for predicting corn price variations from August to December. The similarity of the BSS values obtained with FAPAR and LAI can be explained by the fact that the GLASS Leaf Area Index (LAI) was used as an input in calculating the GLASS FAPAR, as noted in Xiao et al. (2015). The relatively good performances obtained with the spatial average of the yearly difference of July GPP are consistent with the fact that July GPP yearly differences were found to be good predictors of yield variations. Corn yield in the U.S. Corn Belt is primarily influenced by the weather conditions during the summer months of July and August (Tannura et al., 2008; Ray et al., 2015), even though the impact of weather factors can differ significantly across regions due to the influence of other factors such as technology, soil condition, and irrigation (Leng, 2017). The good relationship between GPP values and corn prices in this study aligns with the findings of Zelingher et al. (2021). Their study demonstrated that regional yield variations, particularly in the USA, exert a substantial influence on global corn prices.

When forecasting price variation, the best performances were obtained with the GLM model based on the spatial average of the yearly difference of July GPP when predicting October price variation, as shown in Figs. 5 & 7. In most cases, the BSS was close to zero or negative from February to May (Fig. 5). It then increased and peaked in July, before declining again. This pattern is similar to the pattern of BSS obtained when predicting yield variations (Fig. 2).

The maps in Fig. 6 illustrate the yearly variation of Gross Primary Productivity (GPP) for July 1984 and 1993. In 1984, the GPP values were significantly higher than the previous year, while the 1993 maps

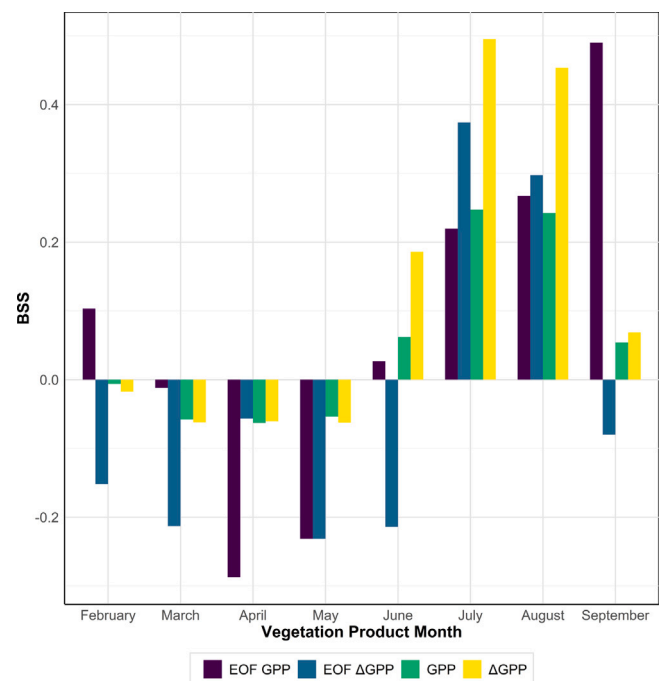


Fig. 5. Brier Skill Score (BSS) assessing predictions of October price variation obtained at different dates, from February to October (start of the harvest period). EOF: predictions based on EOF applied to monthly Gross Primary Production (GPP) available from February to October. GPP: predictions based on spatial averages of monthly GPP computed from February to October. ΔEOF GPP: predictions derived from EOF applied to yearly differences of monthly GPP (month of one year compared to the same month of the previous year). ΔGPP: predictions derived on spatial averages of yearly differences of monthly GPP.

indicate a significant decrease in GPP values. These GPP variations are also reflected in the U.S. yield and production of corn in the United States. After a drought-induced decline in 1983, US corn production experienced a remarkable rebound in 1984, with a production increase of 83% to attain 7.67 billion bushels from 4.17 billion bushels in 1983. The average yield of corn in 1984 also improved significantly, from 81.1 bushels per acre in 1983 to 106.6 bushels per acre. In contrast, the average yield of corn in the US decreased to 100.7 bushels per acre in

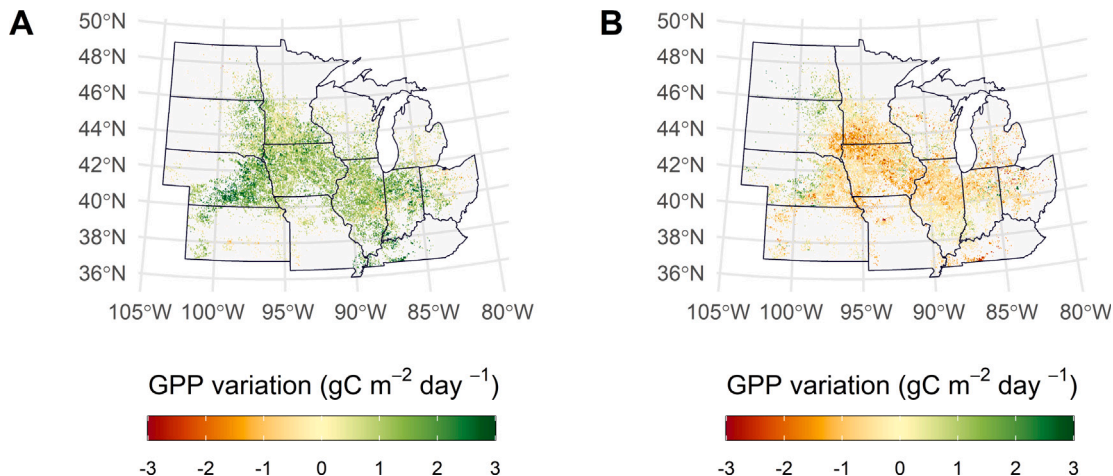


Fig. 6. Map A shows the July GPP differences between 1984 and 1983. Map B illustrates the July GPP differences between 1993 and 1992. The years 1984 was characterized by above-than-average corn production, while 1993 was impacted by the Great Flood leading to reduced crop production across several U.S. states.

1993, a notable drop from the 131.5 bushels per acre achieved in 1992. Similarly, the US corn production for 1993 declined significantly to 6.34 billion bushels, down from the 9.48 billion bushels produced in the previous year. The pronounced variability of yields had a substantial effect on corn prices. In October 1984, for instance, the price of corn was 18% lower than the previous year, dropping from 148.15 to 121.19 USD per metric ton. On the other hand, in October 1993, the corn price increased by 12% from the previous year, rising from 95.11 to 106.53 USD per metric ton (Corn price (U.S.), no. 2, yellow, f.o.b. US Gulf ports).

The GLM, as per Eq. (11), indicates a negative estimate (−3.8) for the parameter β_1 in predicting October's price variation. It means that the chance of a price increase in October is lower when the average yearly variation of the July GPP is positive, i.e., in the case of a July GPP increase compared to the previous year. In other words, an average July GPP higher compared to the previous year increases the chance of a price decrease.

The good performance of the spatial average of the yearly difference of July GPP model is illustrated in Fig. 7.A, where the predicted probabilities of price increase are presented for the years of actual price increase and decrease, separately. Clearly, the predicted probabilities of price increase tend to be higher in case of an actual price increase. The median probability is higher than 0.87 in the case of an actual price increase but is lower than 0.07 in the case of an actual price decrease. The good classification performance of the model is confirmed in Fig. 7.B, where the ROC curve of the model is presented. The AUC reaches 0.91, with a 95% confidence interval of [0.83–1].

The scatter plots (panel C in Fig. 7) provide another visual representation of the relationship between the predicted probabilities of October price increase and the actual October price changes. When the model predicts a probability of a price increase greater than 0.5, most of the actual October price changes are greater than 0, indicating a price increase. Conversely, when the predicted probabilities of price increase are lower than 0.5, it is observed that most of the actual October price changes are lower than 0, indicating a price decrease.

To compare our model with the WASDE report, we limited our analysis to GPP data from 1993 to 2018 because the WASDE yield prediction data were restricted to this time period. Specifically, we compared the performance of our best model for price prediction (GLM with average July GPP variation as a predictor), with another GLM model using the WASDE yield predictions extracted from the July reports as predictors. The output of both types of models is the probability of price increase compared to the previous year, for the months of August to December. Table 2 show several performance metrics for the

Table 2
Performance of two models predicting monthly price changes using either yield forecasts from the July World Agricultural Supply and Demand Estimates (WASDE) report or annual variations in average July Gross Primary Productivity (GPP). The rows correspond to the different price months (August to December) used to evaluate the yearly price changes.
BSS: Brier Skill Score; AUC: Area Under the Curve; MCC: Matthew Correlation Coefficient.

Monthly price	Performance metrics		
	BSS	AUC	MCC
WASDE			
August	0.15	0.68	0.12
September	0.10	0.62	0.12
October	0.26	0.78	0.37
November	0.12	0.68	0.21
December	0.03	0.47	−0.12
Δ GPP			
August	0.03	0.58	0.20
September	0.21	0.76	0.36
October	0.38	0.85	0.52
November	0.13	0.69	0.36
December	−0.02	0.50	0.20

two model types. Performances were poor with both types of models for predicting price variations in December but better for the other months. This is probably because the July GPP and July WASDE reports were not informative anymore to predict the price variations in December. The best results were obtained with the GLM model using July GPP variation as a predictor when predicting the price variations in October. In this case, the BSS was equal to 0.38, the AUC was equal to 0.85, and the MCC reached 0.52. In most cases, the performances were lower with the GLM using WASDE July yield forecasts as the predictor. Fig. 8 shows the predicted price changes in October obtained with the GLM using WASDE July yield forecasts (A) and GLM using average July GPP variations (B) and. The predicted probabilities are shown separately for the years of an actual price increase and the years of an actual price decrease. Clearly, the contrast between the years of actual increase and decrease is stronger with the GLM using average July GPP variations as predictors. Furthermore, relative price changes observed in the data correlate more strongly with average July GPP variations than with WASDE July yield forecasts. These findings confirm the efficacy of the GLM model that utilizes average July GPP variation for predicting corn price fluctuations.

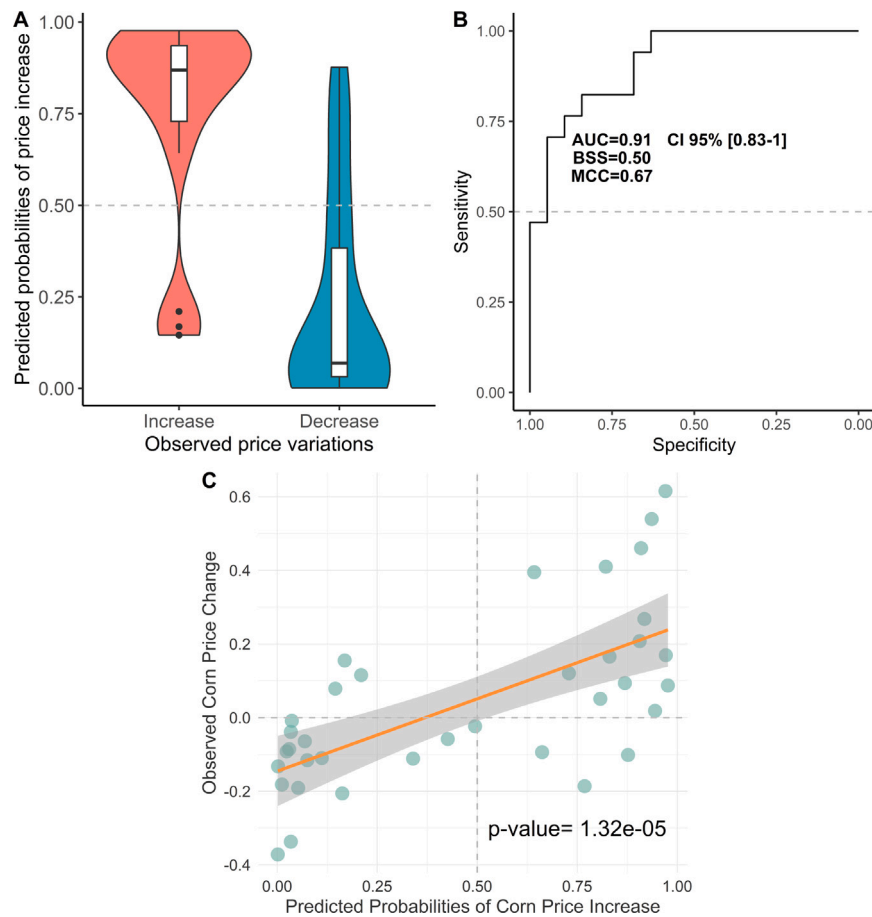


Fig. 7. Boxplots (A) represent predicted probabilities for price increases and decreases. The model's classification performance is shown by the ROC curve (B) with an AUC of 0.91, a BSS of 0.50 and the MCC score of 0.67. The scatter plot (C) illustrates the relationship between predicted probabilities and actual price changes.

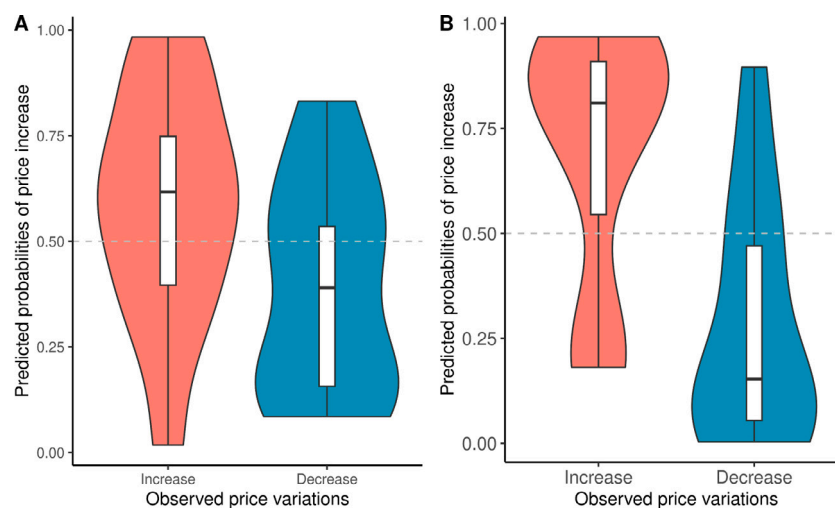


Fig. 8. Boxplots represent the predicted probabilities of a price increase from 1993 to 2018 in case of an actual price increase and in case of an actual price decrease in October. The predictions are obtained using two Generalized Linear Models (GLMs). The model in panel A uses the WASDE July yield forecasts as a predictor, while the model in panel B uses the average July GPP variations as predictors.

4. Conclusion

The present study demonstrates the efficacy of summarizing maps of GPP satellite products using Empirical Orthogonal Function analysis and using resulting principal components to predict US corn yield

variations with high accuracy. Additionally, the study reveals that spatial averages of satellite GPP data obtained during July and August in the US Corn Belt can be leveraged to forecast upcoming fluctuations in global corn prices. These findings suggest that utilizing GPP data yields superior classification performance compared to the current

Table A.1

The table presents the estimated regression parameters of the optimal GLM model for forecasting yearly yield variations. The coefficients for each selected predictor (PC) identified by the LASSO are included in the table, with their corresponding values displayed in the Coefficient column.

Predictor	Coefficient
(Intercept)	0.99
PC_001	3.69
PC_003	0.11
PC_004	−0.82
PC_005	1.36
PC_011	0.34
PC_016	−0.25
PC_017	−1.69
PC_019	−0.44
PC_021	−0.06
PC_025	0.20
PC_026	−0.83
PC_027	0.42
PC_029	−0.53
PC_033	−1.14

WASDE yield forecasting method used in the US. Ultimately, this study underscores the potential of leveraging satellite data maps to effectively forecast crop yield and corn price fluctuations several months prior to harvest.

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CRedit authorship contribution statement

Florian Teste: Designed the research, Performed the research, Analyzed the data, Wrote the paper. **David Makowski:** Designed the research, Performed the research, Wrote the paper. **Hassan Bazzi:** Wrote the paper. **Philippe Ciaï:** Designed the research, Wrote the paper.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Florian TESTE reports a relationship with Atos SE that includes: employment. Hassan Bazzi reports a relationship with Atos SE that includes: employment.

Data availability

All the data used are publicly and freely available. The data sources can be found in references.

Appendix. Appendices

A.1. Model parameters

See [Table A.1](#).

A.2. BSS scores for the yield variation prediction

See [Figs. A.1–A.3](#).

A.3. BSS scores for corn price variation prediction

See [Figs. A.4–A.15](#).

Product month	EOF	EOF ΔFAPAR	Monthly average	Δ Monthly average
February	−0.10	0.06	−0.06	−0.04
March	−0.12	−0.26	−0.04	−0.02
April	−0.11	−0.18	−0.06	−0.05
May	−0.13	−0.23	−0.07	−0.06
June	0.10	−0.10	−0.06	−0.05
July	−0.11	−0.39	−0.07	0.14
August	−0.01	−0.19	−0.06	0.13
September	0.09	0.10	−0.03	0.12
October	0.04	−0.02	−0.04	−0.04

Fig. A.1. BSS scores of corn yield variations prediction based on the FAPAR products. The row represents the monthly GLASS products, while the columns represent the dimension reduction method used.

Product month	EOF	EOF ΔLAI	Monthly average	Δ Monthly average
February	−0.01	−0.07	−0.07	−0.04
March	−0.02	0.01	−0.06	−0.01
April	0.09	−0.22	−0.07	−0.05
May	−0.06	−0.14	−0.07	−0.06
June	−0.08	−0.31	−0.06	−0.06
July	−0.15	−0.36	−0.05	0.12
August	−0.25	−0.23	−0.03	0.11
September	−0.28	−0.26	−0.01	0.11
October	−0.22	−0.15	−0.05	−0.05

Fig. A.2. BSS scores of corn yield variations prediction based on LAI products. The row represents the monthly GLASS products, while the columns represent the dimension reduction method used.

Product month	EOF	EOF ΔGPP	Monthly average	Δ Monthly average
February	−0.21	−0.28	−0.04	−0.06
March	−0.14	−0.14	−0.05	−0.03
April	−0.04	0.34	−0.07	−0.07
May	−0.26	−0.02	−0.06	−0.06
June	0.03	−0.27	0.00	0.15
July	−0.12	0.75	0.07	0.42
August	−0.20	0.46	0.03	0.28
September	−0.18	0.05	0.00	0.19
October	0.02	−0.13	−0.04	−0.04

Fig. A.3. BSS scores of corn yield variations prediction based on GPP products. The row represents the monthly GLASS products, while the columns represent the dimension reduction method used.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	0.06	0.08	0.09	-0.03	-0.06	-0.07	-0.06	-0.06	-0.05	-0.04
March	—	0.16	0.05	-0.05	-0.05	-0.06	-0.06	-0.05	-0.06	-0.06
April	—	—	-0.04	-0.07	0.00	-0.04	-0.03	-0.05	-0.07	-0.07
May	—	—	—	-0.07	-0.01	-0.04	-0.04	-0.05	-0.07	-0.07
June	—	—	—	—	-0.07	-0.06	-0.04	-0.05	-0.05	-0.05
July	—	—	—	—	—	0.12	0.26	0.27	0.25	0.20
August	—	—	—	—	—	—	0.22	0.31	0.33	0.29
September	—	—	—	—	—	—	—	0.20	0.23	0.21
October	—	—	—	—	—	—	—	—	-0.05	-0.04

Fig. A.4. BSS scores of corn price variations prediction based on Δz_{mt}^{FAPAR} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.04	-0.06	-0.05	-0.04	-0.06	-0.06	-0.02	-0.02	-0.03	-0.06
March	—	0.02	0.02	0.01	-0.06	-0.05	-0.05	-0.06	-0.06	-0.04
April	—	—	-0.01	-0.06	0.01	-0.05	-0.04	-0.06	-0.06	-0.06
May	—	—	—	-0.05	0.03	-0.04	-0.06	-0.06	-0.01	-0.02
June	—	—	—	—	0.03	0.00	0.13	0.19	0.17	0.11
July	—	—	—	—	—	0.12	0.38	0.50	0.31	0.17
August	—	—	—	—	—	—	0.28	0.45	0.40	0.45
September	—	—	—	—	—	—	—	0.07	0.11	0.16
October	—	—	—	—	—	—	—	—	0.01	-0.04

Fig. A.5. BSS scores of corn price variations prediction based on Δz_{mt}^{GPP} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	0.09	0.10	0.17	0.01	-0.05	-0.07	-0.06	-0.07	-0.04	-0.03
March	—	0.17	0.08	-0.04	-0.05	-0.06	-0.06	-0.04	-0.06	-0.06
April	—	—	-0.05	-0.06	0.02	-0.02	-0.01	-0.03	-0.06	-0.06
May	—	—	—	-0.07	0.01	-0.02	-0.03	-0.05	-0.07	-0.06
June	—	—	—	—	-0.07	-0.06	-0.04	-0.05	-0.05	-0.05
July	—	—	—	—	—	0.09	0.25	0.25	0.25	0.19
August	—	—	—	—	—	—	0.22	0.30	0.33	0.28
September	—	—	—	—	—	—	—	0.18	0.22	0.19
October	—	—	—	—	—	—	—	—	-0.04	-0.04

Fig. A.6. BSS scores of corn price variations prediction based on Δz_{mt}^{LAI} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.11	-0.06	-0.10	-0.06	0.24	-0.05	-0.12	0.08	-0.03	0.24
March	—	-0.15	-0.08	-0.10	0.09	-0.09	-0.10	-0.07	-0.25	0.01
April	—	—	-0.09	-0.12	0.10	-0.19	-0.20	0.24	0.05	0.18
May	—	—	—	-0.08	-0.12	-0.01	0.05	-0.04	0.10	-0.08
June	—	—	—	—	-0.07	0.03	0.01	-0.08	-0.34	-0.02
July	—	—	—	—	—	-0.03	-0.24	-0.11	-0.19	-0.07
August	—	—	—	—	—	—	-0.09	-0.16	-0.15	-0.11
September	—	—	—	—	—	—	—	-0.02	0.17	-0.13
October	—	—	—	—	—	—	—	—	-0.24	0.20

Fig. A.7. BSS scores of corn price variations prediction based on the EOF of Δz_{smt}^{FAPAR} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	0.13	-0.25	-0.11	-0.06	0.13	-0.11	-0.22	-0.16	-0.13	-0.41
March	—	0.19	-0.23	-0.24	0.08	-0.06	-0.28	-0.10	-0.24	-0.11
April	—	—	0.11	-0.07	-0.12	-0.23	-0.03	0.02	-0.25	-0.01
May	—	—	—	-0.14	-0.17	-0.14	-0.06	0.00	-0.14	-0.10
June	—	—	—	—	-0.31	-0.22	-0.12	0.09	-0.06	-0.09
July	—	—	—	—	—	-0.15	0.03	-0.21	-0.25	-0.11
August	—	—	—	—	—	—	-0.03	-0.15	-0.21	-0.04
September	—	—	—	—	—	—	—	-0.31	0.22	-0.11
October	—	—	—	—	—	—	—	—	0.00	-0.04

Fig. A.8. BSS scores of corn price variations prediction based on the EOF on Δz_{smt}^{LAI} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.19	-0.29	-0.07	-0.32	-0.03	-0.05	-0.13	-0.15	-0.18	-0.04
March	—	0.09	0.23	-0.21	0.30	-0.16	-0.17	-0.21	-0.39	-0.39
April	—	—	0.18	-0.15	-0.14	-0.23	-0.05	-0.06	-0.20	0.00
May	—	—	—	0.01	-0.24	-0.32	0.04	-0.23	-0.35	-0.34
June	—	—	—	—	-0.05	-0.18	-0.35	-0.21	-0.21	-0.20
July	—	—	—	—	—	-0.07	0.34	0.37	0.16	-0.20
August	—	—	—	—	—	—	0.06	0.30	0.25	0.18
September	—	—	—	—	—	—	—	-0.08	0.08	-0.21
October	—	—	—	—	—	—	—	—	-0.07	0.00

Fig. A.9. BSS scores of corn price variations prediction based on the EOF on Δz_{smt}^{GPP} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.03	0.00	-0.05	-0.03	-0.06	-0.07	-0.05	-0.08	0.00	-0.05
March	—	-0.04	0.00	-0.07	-0.07	0.00	0.00	0.00	0.00	-0.01
April	—	—	0.02	-0.10	-0.06	0.03	0.01	0.00	0.00	0.00
May	—	—	—	-0.08	-0.12	-0.01	0.05	-0.04	0.10	-0.08
June	—	—	—	—	-0.04	-0.02	-0.08	-0.04	-0.05	-0.09
July	—	—	—	—	—	-0.02	-0.08	-0.04	-0.05	-0.09
August	—	—	—	—	—	—	-0.08	-0.08	-0.08	-0.08
September	—	—	—	—	—	—	—	0.03	0.02	0.04
October	—	—	—	—	—	—	—	—	0.00	-0.04

Fig. A.10. BSS scores of corn price variations prediction based on the EOF on z_{smt}^{FAPAR} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.03	-0.02	-0.12	-0.07	-0.06	-0.07	0.00	-0.01	-0.04	-0.08
March	—	-0.03	-0.03	-0.02	0.00	-0.05	0.00	0.00	0.00	0.00
April	—	—	-0.02	0.03	-0.05	-0.03	-0.05	0.00	-0.04	0.00
May	—	—	—	0.00	-0.13	-0.07	0.00	-0.05	-0.19	-0.02
June	—	—	—	—	-0.02	-0.01	-0.05	-0.08	-0.07	-0.08
July	—	—	—	—	—	-0.01	-0.05	-0.08	-0.07	-0.08
August	—	—	—	—	—	—	-0.07	-0.17	-0.05	-0.08
September	—	—	—	—	—	—	—	-0.05	0.03	-0.07
October	—	—	—	—	—	—	—	—	-0.01	-0.03

Fig. A.11. BSS scores of corn price variations prediction based on the EOF on z_{smt}^{LAI} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.16	-0.33	-0.39	-0.27	-0.14	-0.06	0.01	0.10	-0.05	-0.03
March	—	-0.13	-0.03	-0.01	-0.06	-0.16	-0.09	-0.01	-0.26	-0.18
April	—	—	-0.01	0.00	-0.21	0.41	0.37	-0.29	0.04	0.00
May	—	—	—	0.01	-0.24	-0.32	0.04	-0.23	-0.35	-0.34
June	—	—	—	—	-0.02	-0.10	-0.11	0.03	0.13	0.06
July	—	—	—	—	—	0.00	-0.26	0.22	-0.25	0.01
August	—	—	—	—	—	—	0.10	0.27	-0.07	-0.01
September	—	—	—	—	—	—	—	0.49	-0.17	-0.26
October	—	—	—	—	—	—	—	—	0.13	-0.23

Fig. A.12. BSS scores of corn price variations prediction based on the EOF on z_{smt}^{GPP} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	0.10	0.13	0.11	0.06	0.08	0.00	-0.04	-0.01	-0.02	-0.01
March	—	0.08	0.01	-0.02	-0.02	-0.03	-0.06	-0.05	-0.05	-0.05
April	—	—	-0.06	-0.06	-0.04	-0.06	-0.05	-0.06	-0.04	-0.04
May	—	—	—	-0.06	-0.03	-0.06	-0.05	-0.04	-0.03	-0.03
June	—	—	—	—	-0.03	-0.02	-0.06	-0.06	-0.06	-0.06
July	—	—	—	—	—	0.08	-0.02	-0.03	-0.04	-0.03
August	—	—	—	—	—	—	0.02	0.05	0.02	0.05
September	—	—	—	—	—	—	—	0.22	0.10	0.16
October	—	—	—	—	—	—	—	—	-0.01	-0.01

Fig. A.13. BSS scores of corn price variations prediction based on z_{mt}^{FAPAR} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	0.14	0.16	0.18	0.12	0.06	0.01	-0.05	-0.01	-0.02	0.00
March	—	0.09	0.05	0.01	-0.03	-0.03	-0.06	-0.04	-0.05	-0.05
April	—	—	-0.06	-0.06	-0.02	-0.06	-0.05	-0.06	-0.05	-0.05
May	—	—	—	-0.06	-0.02	-0.06	-0.05	-0.05	-0.04	-0.04
June	—	—	—	—	-0.03	-0.01	-0.06	-0.06	-0.06	-0.06
July	—	—	—	—	—	0.12	-0.01	-0.01	-0.02	-0.01
August	—	—	—	—	—	—	0.04	0.08	0.05	0.09
September	—	—	—	—	—	—	—	0.22	0.11	0.18
October	—	—	—	—	—	—	—	—	0.01	0.01

Fig. A.14. BSS scores of corn price variations prediction based on z_{mt}^{LAI} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

Monthly product	March	April	May	June	July	August	September	October	November	December
February	-0.02	-0.04	-0.06	-0.05	-0.03	-0.03	-0.04	-0.01	-0.06	-0.06
March	—	0.03	0.00	0.04	-0.05	-0.05	-0.05	-0.06	-0.06	-0.05
April	—	—	-0.03	-0.04	-0.04	-0.06	-0.06	-0.06	-0.06	-0.06
May	—	—	—	-0.06	0.00	-0.06	-0.06	-0.05	-0.05	-0.06
June	—	—	—	—	0.05	0.06	0.01	0.06	0.02	0.04
July	—	—	—	—	—	0.12	0.15	0.25	0.13	0.12
August	—	—	—	—	—	—	0.15	0.24	0.16	0.24
September	—	—	—	—	—	—	—	0.05	0.06	0.13
October	—	—	—	—	—	—	—	—	0.10	0.08

Fig. A.15. BSS scores of corn price variations prediction based on z_{mt}^{GPP} . The row represents the monthly GLASS products, and the columns represent the monthly corn price variation that we intend to forecast.

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